COL 703/7203

October 29th

A formula is in prenex form if it is written as

$$Q_1 y_1 \ldots Q_n y_n \; \phi$$

where each $Q_i \in \{\exists, \forall\}$

$n \geq 0$

ϕ contains no quantifiers

also called the "matrix" of the entire formula

If this is in CNF, then the formula is said to be in PCNF form.

$$\neg\big(\exists x\, P(x,y) \wedge \forall z\, Q(z)\big) \vee \exists w\, Q(w)$$

$$\big(\neg\exists x\, P(x,y) \vee \neg\forall z\, Q(z)\big) \wedge \exists w\, Q(w)$$

$$\forall x\, \neg P(x,y) \vee \exists z\, \neg Q(z) \wedge \exists w\, Q(w)$$

$$\forall x\, \exists z\, \exists w \Big(\neg P(x,y) \vee \neg Q(z) \vee Q(w)\Big)$$

Translation Lemma

If t is a term and ϕ is a formula such that t is free for x in ϕ, then

$$A \models \phi[t/x] \quad \text{iff} \quad A_{[x \to A[\![t]\!]]} \models \phi$$

Example

$\phi = \forall y\, P(x)$

This is to illustrate that "t is free for x in ϕ" is crucial.

If t is y (y is not free for x in ϕ)

$\phi[t/x] = \forall y\, P(y)$ ← there is a variable capture that happens which changes the meaning of the formula

If t is z

$\phi[t/x] = \forall y\, P(z)$

Claim Let $\phi = Qx\,\psi$ be a formula where $Q \in \{\exists, \forall\}$.

Let y be a variable that does not occur in ψ.

Then $\phi \equiv Qy\,\psi[y/x]$

$A \models \exists y\,\psi[y/x]$

iff $A_{[y \mapsto a]} \models \psi[y/x]$ for some $a \in U_A$

$$\mathcal{A}_{[y \mapsto a]} \models \psi[y/x]$$

iff

$$\mathcal{A}_{[y \mapsto a]}[x \mapsto \mathcal{A}_{[y \mapsto a]}[\![y]\!]] \models \psi$$

iff $\mathcal{A}_{[y \mapsto a][x \mapsto a]} \models \psi$

iff $\mathcal{A}_{[a \mapsto a]} \models \psi$

iff $\mathcal{A} \models \exists x \psi$

Proved.

Translation Lemma

$\mathcal{A} \models \phi[y/x]$

iff $\mathcal{A}_{[x \mapsto \mathcal{A}[\![y]\!]]} \models \phi$

for some $a \in U_{\mathcal{A}}$

for some $a \in U_{\mathcal{A}}$

for some $a \in U_{\mathcal{A}}$

Relevance Lemma

↳ y does not occur in ψ and therefore not relevant

A formula is rectified if no variable occurs both bound and free, and every quantifier refers to a different variable.

$$\forall x \exists y \; P(x, f(y)) \wedge \forall y \; (Q(x,y) \wedge R(x))$$

$$\forall u \exists v \; P(u, f(v)) \wedge \forall y \; (Q(x,y) \wedge R(x))$$

Clarification: Note that rectification is done by simply renaming the bound variables. Therefore the rectified formula is logically equivalent to the original formula. (The assignment does not talk about bound variables anyway!)

There was another clarification question that had come up at this point.

$\forall x\ P(x) \wedge Q(x)$ $\qquad$ $\forall y\ P(y) \wedge Q(y)$

Are these two formulas logically equivalent?

Clearly, not. Consider the assignment $\mathcal{A}$ where $U_{\mathcal{A}} = \mathbb{N}$, $P_{\mathcal{A}} = \mathbb{N}$, $Q_{\mathcal{A}} = \{3\}$, $x = 3$, $y = 4$.

This satisfies the formula on the left but not the one on the right.

Are they equisatisfiable? → Of course. Map x to whatever y was being mapped to (or the other way around).

But just telling that they are equisatisfiable is a weak statement. We know much better — the formulas are structurally the same and therefore it is trivial to produce a satisfying assignment of one from that of the other one.

↳ And this is what the translation lemma tells us.

A formula in rectified prenex form is in Skolem form if it does not contain any existential quantifiers.

$$\forall x \exists y \; P(x,y) \qquad \forall x \; P(x, f(x))$$

equisatisfiable.

Idea: Treat f as the "selection function" which gives one such y for every x.

$\forall x \; \exists y \; P(x,y)$

$\forall x \; P(x, f(x))$

Consider the assignment

$\mathbb{Z}$, $<$

$P(x,y)$ is true if $x<y$.

$f(x) = x-1$

Skolemization

$$\forall x \left(P(x) \to Q(x) \right) \to \left(\forall x\, P(x) \to \forall x\, Q(x) \right)$$

Clarification question: why can't the quantifiers be pulled out at this point itself?

$$\boxed{\forall x \left(P(x) \to Q(x) \right) \to \left(\forall y\, P(y) \to \forall z\, Q(z) \right)}$$

$$\neg \forall x \left(P(x) \to Q(x) \right) \vee \left(\forall y\, P(y) \to \forall z\, Q(z) \right)$$

$$\exists x \neg \left(P(x) \to Q(x) \right) \vee \left(\neg \forall y\, P(y) \vee \forall z\, Q(z) \right)$$

$$\exists x \neg \left(\neg P(x) \vee Q(x) \right) \vee \left(\exists y \neg P(y) \vee \forall z\, Q(z) \right)$$

$$\exists x \left(P(x) \wedge \neg Q(x) \right) \vee \exists y \neg P(y) \vee \forall z\, Q(z)$$

See the answer on the last slide.

$\exists x \exists y \forall z \left((P(x) \wedge \neg Q(x)) \vee \neg P(y) \vee Q(z) \right)$

$(a \wedge b) \vee c$
$(a \vee c) \wedge (b \vee c)$

prenex

PCNF

$\forall z$ $(P(a) \wedge \neg Q(a)) \vee (\neg P(b)) \vee Q(z)$

$(P(x) \vee \neg P(y) \vee Q(z)) \wedge (\neg Q(x) \vee \neg P(y) \vee Q(z))$

$\{P(x), \neg P(y), Q(z)\}$, $\{\neg Q(x), \neg P(y), Q(z)\}$

$$\forall z \; \exists x \; \exists y \; \Big(\big(P(x) \wedge \neg Q(x) \big) \vee \neg P(y) \vee Q(z) \Big)$$

$$\forall z \Big(\big(P(f(z)) \wedge \neg Q(f(z)) \big) \vee \neg P(g(z)) \vee Q(z) \Big)$$

The quantifiers are extracted out in an "outermost-first" manner.

$$\exists x \forall y \; P(x,y) \rightarrow \forall y \exists x \; P(x,y)$$

$$\exists x \forall y \; P(x,y) \rightarrow \forall z \exists w \; P(w,z)$$

$$\neg \exists x \forall y P(x,y) \vee \forall z \exists w \; P(w,z)$$

$$\forall x \exists y \neg P(x,y) \vee \forall z \exists w \; P(w,z)$$

$$\forall x \exists y \forall z \exists w \; (\neg P(x,y) \vee P(w,z))$$

$$\forall x \forall z \exists w \; (\neg P(x,f(x)) \vee P(w,z))$$

$$\forall x \forall z \; (\neg P(x,f(x)) \vee P(g(x,z),z))$$

Clarification

$\forall x (P(x) \rightarrow Q(x)) \rightarrow (\forall y P(y) \rightarrow \forall z Q(z))$

Why can't the quantifiers be extracted out at this point itself?

↳ It is because $\rightarrow$ hides a negation in it. ($a \rightarrow b$ is $\neg a \vee b$)

And pulling quantifiers over a negation changes the quantifier.

Therefore we first push all the negations inside and then pull the quantifiers out.